

QuaRK: A Resource-Aware Quantum Reservoir Kernel for Temporal Learning

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Abstract

Learning from temporal data requires a representation that retains useful history while controlling statistical and computational complexity. We introduce QUARK, a quantum-reservoir-based temporal learning framework that combines a finite-set Johnson–Lindenstrauss projection, independently parameterized reset-contractive quantum reservoirs, local-observable features, and a regularized Matérn kernel readout. The analysis separates four properties that are sometimes conflated in quantum reservoir learning. First, the reset channel yields exact trace-norm contraction and a geometrically fading dependence on initialization. Second, distinct exact feature vectors can be interpolated by a finite-norm Matérn RKHS readout, a conditional capacity statement that does not assert injectivity of the quantum feature map. Third, local Pauli observables admit a classical-shadow measurement guarantee whose finite-shot error can be propagated to a fixed readout. Fourth, for a fixed exact featurizer and stationary beta-mixing input–output data, an independent-block argument gives a one-sided risk bound with an explicit effective block count and a finite-window truncation term. These results provide a resource-aware and assumption-explicit foundation for evaluating quantum reservoir kernels without asserting quantum advantage, hardware-noise robustness, or superiority to classical temporal models.

1 Introduction

Temporal prediction differs from ordinary supervised learning because a useful representation must retain information from a potentially unbounded past. Reservoir computing addresses this problem by evolving a fixed dynamical system and training only a comparatively simple readout (Jaeger, 2001; Maass et al., 2002). Contractive reservoirs are particularly attractive: contraction makes the effect of initialization vanish, produces a well-defined causal filter, and gives a direct handle on fading memory. Quantum reservoir computing instantiates the dynamical system with a quantum channel and extracts classical features from the resulting states (Fujii & Nakajima, 2017; Chen et al., 2020). This construction can provide structured nonlinear temporal features, but it also introduces resource questions that are absent from an abstract reservoir model: how many qubits encode an input, how many reservoir instances are used, which observables are read out, and how many state preparations are required to estimate them.

We study these questions through QUARK, a quantum-reservoir kernel for temporal learning. A finite-set Johnson–Lindenstrauss (JL) map compresses each input vector to the width of a quantum register. A collection of independently parameterized reset-contractive channels processes a window sequentially. Expectations of local observables from the resulting states are concatenated into a classical feature vector, and a Matérn kernel ridge-regression readout is fitted to these features. The resulting learner is modular: the projection controls encoded width, contraction controls memory, multiplexing and the observable family control feature dimension, and RKHS regularization controls the readout class.

The purpose of the present analysis is not to infer a quantum advantage from this modular construction. Instead, it identifies what follows from each module and what does not. In particular, finite-set distance preservation by JL is not feature injectivity; the ability of a strictly positive-definite kernel to interpolate distinct features is not a guarantee that the reservoir produces distinct features; and finite-shot estimation

of local observables is not a device-noise analysis. Keeping these statements separate is necessary for a falsifiable account of the method.

Contributions. Our contributions are as follows.

1. We formulate a resource-explicit temporal learner built from reset-contractive quantum reservoirs and a classical kernel readout. Spatial multiplexing is represented as a tuple of independently evolved states and a concatenated observable vector, so the model definition does not rely on a fictitious tensor-product contraction.
2. We prove exact trace-norm contraction of every reset channel and derive uniqueness of the causal state, loss of initialization dependence, and a finite-window feature error that decays as $\sqrt{\sum_r \lambda_r^{2w}}$. A Matérn RKHS stability lemma propagates this error to predictions and squared loss.
3. We characterize finite-sample readout capacity. Conditional on pairwise-distinct exact features, strict positive definiteness of the Matérn kernel gives a finite-norm interpolant and an explicit upper bound on the best training error below the interpolation norm. This result makes no claim that the JL projection or quantum reservoir is injective.
4. We give a one-sided risk bound for a fixed exact featurizer under a stationary beta-mixing input-output process. The proof retains the block length in the independent-block argument of [Mohri & Rostamizadeh \(2008\)](#), exposes the effective number of blocks, and adds the finite-window approximation term. We also state the conditions under which geometric mixing yields a $\sqrt{\log N/N}$ -type estimation rate.
5. We state the local-Pauli classical-shadow requirement separately from the learning bound and give a deterministic perturbation guarantee for evaluating a fixed RKHS readout on estimated rather than exact features.

Together, these results position QUARK as a resource-aware quantum-reservoir-based feature map with an analyzable classical readout. They do not establish computational speedup, statistical superiority over classical reservoirs, robustness to device noise, or practical deployability on current hardware.

Organization. Section 2 relates the construction to reservoir computing, quantum temporal models, and dependent-data learning. Section 3 fixes the stochastic and learning notation. Section 4 defines the complete pipeline and its resources. Section 5 presents the theoretical results, with proofs in Appendix B. Section 6 is deliberately reserved for the empirical revision. Sections 7 and 8 discuss scope, limitations, and conclusions.

2 Related Work

Reservoir computing and fading memory. Echo-state networks and liquid-state machines introduced the principle of driving a fixed recurrent system and training only its readout ([Jaeger, 2001](#); [Maass et al., 2002](#)). Fading memory formalizes continuity of a causal filter with respect to increasingly remote inputs ([Boyd & Chua, 1985](#)). Contractive state updates provide a direct sufficient condition for the echo-state and fading-memory properties, while universality and risk analyses study different questions: the functions representable by families of reservoirs and the statistical behavior of their learned readouts, respectively ([Grigoryeva & Ortega, 2018](#); [Gonon et al., 2020](#)). Our contraction result concerns one fixed reset-based architecture and is not a universality theorem.

Quantum reservoir computing. Quantum reservoir computing uses quantum dynamics as the driven state system ([Fujii & Nakajima, 2017](#)). Existing proposals study dissipative dynamics, reset mechanisms, noisy processors, weak or repeated measurements, and spatial multiplexing ([Chen & Nurdin, 2019](#); [Chen et al., 2020](#); [Nakajima et al., 2019](#); [Molteni et al., 2023](#); [Mujal et al., 2023](#); [Hu et al., 2024](#)). Finite-dimensional quantum reservoir filters and the role of input dependence have also received formal treatment ([Martínez-Peña & Ortega, 2023](#)). QUARK uses a reset channel because its contraction coefficient is exact and directly tunable.

Multiple reservoir instances enlarge the classical observable vector, but they may be executed sequentially on reused hardware; consequently, the instance count is not automatically a peak-width requirement.

Quantum features and kernel readouts. Quantum feature methods encode classical data into states or measurement statistics before applying a classical learner (Schuld & Killoran, 2019; Huang et al., 2021). QUARK belongs to this broad family, with the important difference that its features are generated dynamically from a temporal window rather than independently from one static input. The Matérn readout is an ordinary classical positive definite kernel on the measured feature vectors. Its role here is both algorithmic—kernel ridge regression provides a regularized nonlinear readout—and analytic—the normalized kernel bounds predictions in an RKHS ball and its spectral second moment controls sensitivity to feature perturbations (Genton, 2001; Porcu et al., 2024; Hofmann et al., 2008). Calling the complete pipeline quantum-reservoir-based does not imply that its kernel has no comparable classical construction.

Random projection and measurement. The JL lemma preserves the geometry of a finite point set under a random linear projection whose target dimension depends logarithmically on the set size (Dasgupta & Gupta, 2003; Woodruff, 2014). We use this result to account for the width needed to encode the observed finite set. It leaves the classical cost of multiplying by the projection matrix and the subsequent dynamics untouched. For measurement, classical shadows can estimate many observables from shared randomized measurements (Huang et al., 2020). Under local random Pauli measurements, a Pauli string of weight at most k has a shadow-norm-squared scaling of 3^k ; this is the factor used in our resource statement.

Learning with dependent observations. Independent-block arguments compare a beta-mixing sequence to independent blocks and yield uniform learning bounds in terms of an effective block count (Yu, 1994; Mohri & Rostamizadeh, 2008; Dedecker et al., 2007). Our result applies this machinery to a process of strided temporal windows. It differs from an i.i.d. bound in two linked ways: the confidence condition contains the mixing coefficient at the block separation, and the stochastic terms scale with the number of blocks rather than the raw number of windows. The analysis is conditional on a fixed featurizer and does not cover data-dependent selection among reservoir or kernel configurations.

3 Problem Setting

3.1 Temporal data and histories

Let $Z = (Z_t)_{t \in \mathbb{Z}}$, with $Z_t = (X_t, Y_t)$, be a stationary stochastic process on $\mathbb{R}^d \times \mathbb{R}$. We write $x_{-\infty:0} = (\dots, x_{-1}, x_0)$ for a semi-infinite input history and

$$x_{-w+1:0} = (x_{-w+1}, \dots, x_0) \in (\mathbb{R}^d)^w$$

for its last w inputs. A temporal predictor maps a history to a scalar. The target is not assumed to have finite memory; finite windows are instead an implementable approximation whose error will be controlled through reservoir contraction.

For a window length $w \geq 1$ and integer stride $s = w + g$, where $g \geq 0$, define the stationary window process

$$W_j := (X_{js-w+1:js}, Y_{js}), \quad j \in \mathbb{Z}. \quad (1)$$

The observed sample is $\mathcal{D}_N = (W_1, \dots, W_N)$. Windows need not be independent, even when they do not overlap.

3.2 Quantum-reservoir features

There are R independently parameterized sub-reservoirs. Sub-reservoir r has an n -qubit state $\rho_t^{(r)}$ and an input-dependent channel $T_{r,x}$. For a history, its causal state at time zero is denoted $\rho_{\infty,0}^{(r)}(x_{-\infty:0})$. For a finite window, we initialize the channel at time $-w$ in a fixed state $\rho_{\text{init}}^{(r)}$ and denote its state after consuming the window by $\rho_{w,0}^{(r)}(x_{-w+1:0})$.

Let $\mathcal{O}$ be a finite family of Hermitian observables satisfying $\|O\|_\infty \leq 1$. The exact infinite-history and finite-window feature maps are

$$\Phi_\infty(x_{-\infty:0}) := (\text{Tr}[O\rho_{\infty,0}^{(r)}])_{r \in [R], O \in \mathcal{O}}, \quad (2)$$

$$\Phi_w(x_{-w+1:0}) := (\text{Tr}[O\rho_{w,0}^{(r)}])_{r \in [R], O \in \mathcal{O}} \in \mathbb{R}^p, \quad p = R|\mathcal{O}|. \quad (3)$$

These are concatenations of expectations from a tuple of states. No tensor-product state is needed to define the learner or analyze its memory. An estimated feature vector $\hat{\Phi}_w$ obtained with a finite number of measurements is a separate random object.

3.3 Kernel readout and risks

Let $\kappa : \mathbb{R}^p \times \mathbb{R}^p \rightarrow \mathbb{R}$ be a normalized Matérn kernel and $\mathcal{H}_\kappa$ its RKHS. For a radius $\Lambda > 0$, define

$$\mathcal{H}_{\kappa,\Lambda} := \{h \in \mathcal{H}_\kappa : \|h\|_{\mathcal{H}_\kappa} \leq \Lambda\}.$$

Normalization means $\kappa(z, z) = 1$, so the reproducing property gives

$$|h(z)| \leq \|h\|_{\mathcal{H}_\kappa} \sqrt{\kappa(z, z)} \leq \Lambda \quad \text{for every } h \in \mathcal{H}_{\kappa,\Lambda}. \quad (4)$$

For squared loss $\ell(\hat{y}, y) = (\hat{y} - y)^2$, define the infinite-history risk

$$\mathcal{R}_\infty(h) := \mathbb{E}[\ell(h(\Phi_\infty(X_{-\infty:0})), Y_0)], \quad (5)$$

the finite-window risk

$$\mathcal{R}_w(h) := \mathbb{E}[\ell(h(\Phi_w(X_{-w+1:0})), Y_0)], \quad (6)$$

and the empirical finite-window risk

$$\hat{\mathcal{R}}_{N,w}(h) := \frac{1}{N} \sum_{j=1}^N \ell(h(\Phi_w(X_{js-w+1:js})), Y_{js}). \quad (7)$$

The theoretical learning result compares $\mathcal{R}_\infty$ with $\hat{\mathcal{R}}_{N,w}$. It therefore has both a statistical component $\mathcal{R}_w - \hat{\mathcal{R}}_{N,w}$ and a deterministic window-truncation component $\mathcal{R}_\infty - \mathcal{R}_w$.

3.4 Dependence and fixed-design scope

For sub-sigma-algebras $\mathcal{A}, \mathcal{B}$, write $\beta(\mathcal{A}, \mathcal{B})$ for their absolute-regularity coefficient. We use the convention

$$\beta_Z(q) := \sup_{t \in \mathbb{Z}} \beta(\sigma(Z_u : u \leq t), \sigma(Z_u : u \geq t + q)), \quad q \geq 0. \quad (8)$$

The process is beta mixing if $\beta_Z(q) \rightarrow 0$.

All learning guarantees below condition on the projection, reservoir parameters, observable family, Matérn parameters, window length, stride, and initialization. These objects must be fixed independently of the sample to which the theorem is applied. Uniformity is over the RKHS ball $\mathcal{H}_{\kappa,\Lambda}$, not over data-dependent architecture or hyperparameter choices.

4 Method

4.1 Finite-set projection and encoding width

Let $\mathcal{P} \subset \mathbb{R}^d$ be the set of all time-point vectors appearing in the finite collection of windows to be encoded; $|\mathcal{P}| \leq wN$. For a chosen register width n , draw

$$\Pi \in \mathbb{R}^{n \times d}, \quad \Pi_{ij} \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, 1/n),$$

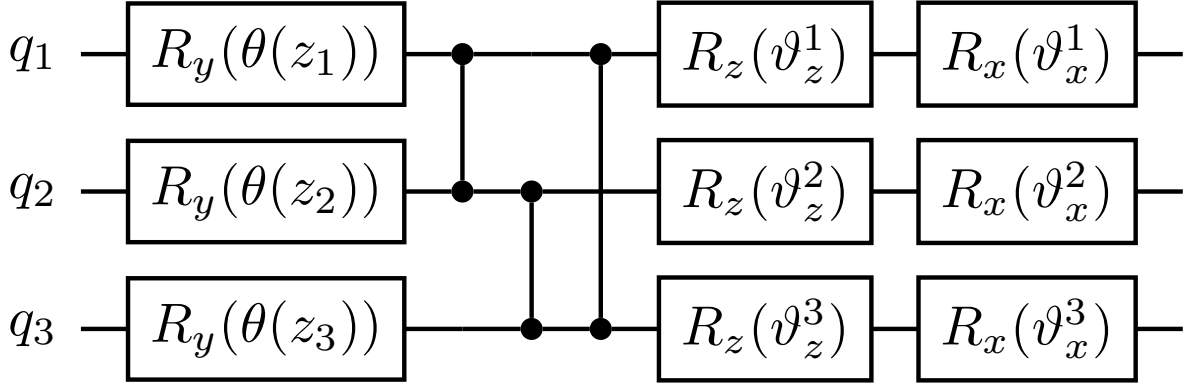


Figure 1: Example input-dependent unitary for a three-qubit sub-reservoir. The circuit is an architectural instance, not a claim about a specific device’s native gate set or noise.

and set $z = \Pi x$. Each coordinate is converted to a bounded rotation angle, for example $\theta(z_j) = \pi \tanh(z_j)$, and encoded with a single-qubit $R_y(\theta(z_j))$ gate.

The JL guarantee in Proposition 2 concerns only the finite point set $\mathcal{P}$. It permits n to be chosen from a distortion and failure probability target instead of setting $n = d$. Applying the dense projection still costs $O(nd)$ arithmetic operations per time point, and neither the bounded angle map nor the reservoir dynamics is asserted to preserve the JL geometry.

4.2 Reset-contractive quantum dynamics

Sub-reservoir r uses the input-dependent unitary

$$V_r(x) = W_r \bigotimes_{j=1}^n R_y(\theta((\Pi x)_j)), \quad (9)$$

where W_r is a fixed hardware-topology-aware unitary. One possible block is an Ising-style layer of edge-wise R_{zz} gates followed by local rotations. Its parameters are sampled once and then held fixed.

After the unitary, apply the reset channel

$$T_{r,x}(\rho) := \lambda_r V_r(x) \rho V_r(x)^\dagger + (1 - \lambda_r) \rho_{\text{reset}}, \quad 0 \leq \lambda_r < 1. \quad (10)$$

The contraction parameter λ_r controls the loss of initialization dependence: larger values retain information longer, while smaller values forget more rapidly. The displayed channel can be dilated with an n -qubit reset register and one coin qubit. For the dilation in Figure 2, $\rho_{\text{reset}} = |+\rangle\langle+|^{\otimes n}$. After the controlled swaps, the ancillary and coin registers are discarded; this partial trace is essential to obtaining the reduced mixture in Equation (10).

For a finite window, the recursion starts from a fixed $\rho_{\text{init}}^{(r)}$ and applies Equation (10) once per time point. The infinite-history state is the limit obtained by moving this initialization into the remote past. Existence and uniqueness of that limit follow from Lemma 3.

4.3 Multiplexing and exact observable features

The R sub-reservoirs are separate channel realizations:

$$(\rho_{w,0}^{(1)}, \dots, \rho_{w,0}^{(R)}).$$

They can be run in parallel on separate registers or sequentially by reusing the same physical register. Parallel execution multiplies peak quantum width; sequential reuse multiplies state preparations and runtime but leaves peak reservoir width unchanged.

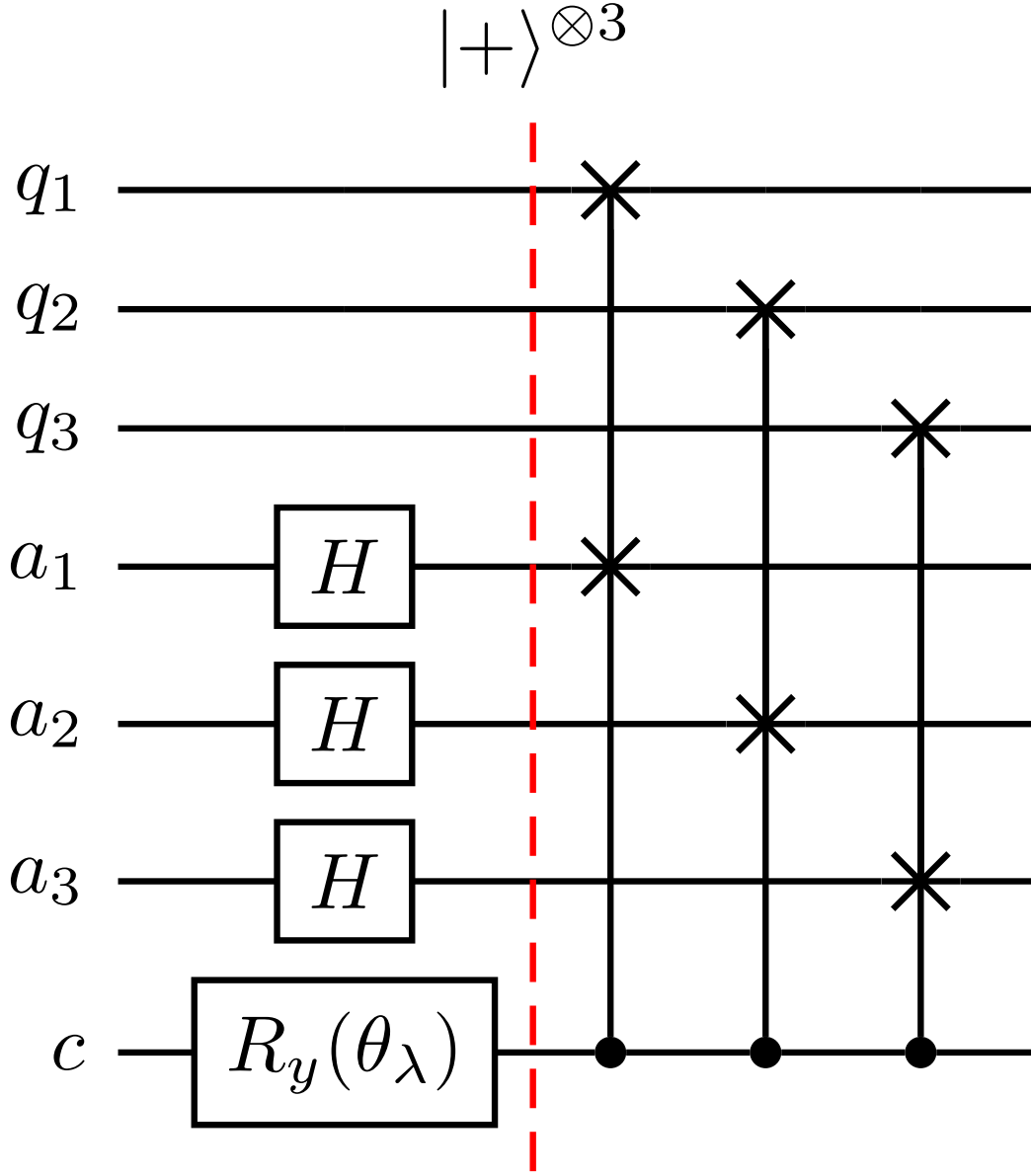


Figure 2: A dilation of the reset channel for three reservoir qubits. Here $\theta_\lambda = 2 \arcsin \sqrt{1 - \lambda}$. The reduced reservoir channel is obtained after discarding the coin and ancilla registers.

For a family $\mathcal{O}$ of local Hermitian observables, the analytic feature map is the concatenation in Equation (3). Its dimension is $p = R|\mathcal{O}|$. Exact expectations are used to define the kernel and in the main generalization theorem. Finite-shot estimates are considered separately in Proposition 12 and Corollary 13.

4.4 Matérn kernel readout

For $r \geq 0$, smoothness $\nu > 1$, and length scale $\xi > 0$, define

$$\varphi_{\nu, \xi}(r) := \frac{2^{1-\nu}}{\Gamma(\nu)} \left(\frac{\sqrt{2\nu} r}{\xi} \right)^\nu K_\nu \left(\frac{\sqrt{2\nu} r}{\xi} \right), \quad (11)$$

Table 1: Principal resource parameters of QUARK.

Quantity	Accounting
Encoded reservoir width	n qubits per sub-reservoir
Displayed reset dilation	Up to $2n + 1$ qubits when reset ancillas and coin are simultaneous
Reservoir realizations	R ; sequential reuse or R -fold parallel width
Exact feature dimension	$p = R \mathcal{O} $
Projection work	$O(nd)$ per time point for a dense Π
Finite-shot work	R state preparations per window per snapshot count; see Proposition 12
Kernel storage/training	$O(N^2)$ memory and $O(N^3)$ naive time

where K_ν is the modified Bessel function of the second kind. Set $\varphi_{\nu,\xi}(0) = 1$, and define

$$\kappa(z, z') := \varphi_{\nu,\xi}(\|z - z'\|_2). \quad (12)$$

The kernel is classical and is evaluated after measurement. Matérn smoothness is not invoked as evidence of quantum expressivity; the properties used below are strict positive definiteness and an RKHS section-stability bound.

Given exact training features $z_i = \Phi_w(x_i)$, kernel ridge regression solves

$$\min_{h \in \mathcal{H}_\kappa} \frac{1}{N} \sum_{i=1}^N (h(z_i) - y_i)^2 + \eta \|h\|_{\mathcal{H}_\kappa}^2. \quad (13)$$

By the representer theorem (Schölkopf & Smola, 2002), the solution is $h(\cdot) = \sum_i \alpha_i \kappa(z_i, \cdot)$, with

$$\boldsymbol{\alpha} = (K + N\eta I)^{-1} \mathbf{y}.$$

Naive dense training requires $O(N^2)$ kernel storage and $O(N^3)$ linear-algebra time. These classical costs remain part of the end-to-end method.

4.5 Resource summary

Table 1 distinguishes peak width, feature dimension, and repeated executions. Big- O entries describe the stated implementation, not an optimality or advantage result.

5 Theory

This section separates the geometric, dynamical, approximation, statistical, and measurement statements used to analyze QUARK. Proofs are in Appendix B.

5.1 Standing assumptions

Assumption 1 (Fixed design and bounded observations). The projection Π , channel parameters, initialization states, observable family $\mathcal{O}$, window parameters w, s , and Matérn parameters (ν, ξ) are fixed independently of the analyzed sample. The observables satisfy $\|O\|_\infty \leq 1$, the labels satisfy $|Y_t| \leq Y$ almost surely, and the Matérn kernel is normalized. The main risk bound uses exact expectations and $\nu > 1$.

The fixed-design condition allows the theorem to hold uniformly over $h \in \mathcal{H}_{\kappa,\Lambda}$. It does not provide uniform control over projections, reservoirs, kernels, or hyperparameters selected using the same observations.

5.2 Finite-set projection

Proposition 2 (Finite-set JL projection). *Let $\mathcal{P} \subset \mathbb{R}^d$ contain $M \geq 2$ points. For $\varepsilon, \delta \in (0, 1)$, there is a universal constant $C > 0$ such that, if*

$$n \geq C\varepsilon^{-2} \log\left(\frac{M}{\delta}\right)$$

and $\Pi_{ij} \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, 1/n)$, then with probability at least $1 - \delta$,

$$(1 - \varepsilon) \|u - v\|_2^2 \leq \|\Pi u - \Pi v\|_2^2 \leq (1 + \varepsilon) \|u - v\|_2^2$$

simultaneously for all $u, v \in \mathcal{P}$. For the points occurring in N windows of length w , one may take $M \leq wN$.

What this establishes. For a specified finite set and distortion target, the number of encoded coordinates—and hence the logical reservoir width in the stated implementation—need only grow logarithmically with the number of points.

What this does not establish. The proposition does not make the angle encoding or reservoir feature map injective, does not imply improved prediction, and does not remove the d -dependent classical cost of applying Π .

5.3 Contraction, causal states, and finite windows

Lemma 3 (Exact trace-norm contraction). *For every sub-reservoir r , input x , and pair of density operators ρ, ρ' ,*

$$\|T_{r,x}(\rho) - T_{r,x}(\rho')\|_1 = \lambda_r \|\rho - \rho'\|_1. \quad (14)$$

Consequently, for each input history there is a unique causal state obtained as the initialization time tends to $-\infty$. Two trajectories driven by the same last w inputs satisfy

$$\left\| \rho_0^{(r)} - \rho_0'^{(r)} \right\|_1 \leq 2\lambda_r^w. \quad (15)$$

What this establishes. The state recursion has the echo-state property in the sense of a unique input-determined causal solution, and initialization dependence vanishes geometrically in trace norm.

What this does not establish. Contraction alone does not imply that different input histories produce different states, that a larger λ_r is more accurate, or that the chosen family of reservoirs is universal.

Proposition 4 (Finite-window feature error). *Under assumption 1, for every input history,*

$$\|\Phi_\infty(x_{-\infty:0}) - \Phi_w(x_{-w+1:0})\|_2 \leq 2\sqrt{|\mathcal{O}| \sum_{r=1}^R \lambda_r^{2w}} \quad (16)$$

$$\leq 2\sqrt{R|\mathcal{O}|} \lambda_{\max}^w, \quad \lambda_{\max} := \max_r \lambda_r. \quad (17)$$

What this establishes. The exact concatenated observable vector of a finite-window recursion approaches the feature vector of the causal infinite-history state at an explicit rate.

What this does not establish. The estimate is a worst-case initialization bound. It need not be tight, and it does not include finite shots or physical noise.

5.4 Matérn stability and truncation of loss

Lemma 5 (Matérn RKHS section stability). *Let κ be the normalized Matérn kernel in Equations (11) and (12) with $\nu > 1$. Then*

$$\|\kappa(z, \cdot) - \kappa(z', \cdot)\|_{\mathcal{H}_\kappa} \leq \frac{1}{\xi} \sqrt{\frac{\nu}{\nu - 1}} \|z - z'\|_2. \quad (18)$$

Corollary 6 (Prediction and loss truncation). *For every $h \in \mathcal{H}_{\kappa, \Lambda}$,*

$$|h(\Phi_\infty) - h(\Phi_w)| \leq \frac{2\Lambda}{\xi} \sqrt{\frac{\nu|\mathcal{O}|}{\nu-1} \sum_{r=1}^R \lambda_r^{2w}}, \quad (19)$$

$$|\mathcal{R}_\infty(h) - \mathcal{R}_w(h)| \leq \tau_w, \quad (20)$$

where

$$\tau_w := \frac{4\Lambda(\Lambda + Y)}{\xi} \sqrt{\frac{\nu|\mathcal{O}|}{\nu-1} \sum_{r=1}^R \lambda_r^{2w}}. \quad (21)$$

What this establishes. The dynamical forgetting rate propagates through every fixed readout in the RKHS ball and through bounded squared loss.

What this does not establish. The result is not an excess-risk or consistency theorem. In particular, for a fixed w , τ_w remains a nonzero upper bound as the sample size grows.

5.5 Conditional interpolation capacity

Theorem 7 (Conditional finite-sample interpolation). *Let $z_i = \Phi_w(x_i) \in \mathbb{R}^p$, $i \in [N]$, be pairwise distinct exact feature vectors, and let $K_{ij} = \kappa(z_i, z_j)$. The Matérn Gram matrix K is positive definite. If $\mathbf{y} \neq 0$, define*

$$\Lambda_\star := \sqrt{\mathbf{y}^\top K^{-1} \mathbf{y}}.$$

Then there is an $h_\star \in \mathcal{H}_\kappa$ with $\|h_\star\|_{\mathcal{H}_\kappa} = \Lambda_\star$ and $h_\star(z_i) = y_i$ for all i . Therefore:

1. *for $\Lambda \geq \Lambda_\star$,*

$$\min_{\|h\|_{\mathcal{H}_\kappa} \leq \Lambda} \widehat{\mathcal{R}}_{N,w}(h) = 0;$$

2. *for $0 \leq \Lambda \leq \Lambda_\star$,*

$$\min_{\|h\|_{\mathcal{H}_\kappa} \leq \Lambda} \widehat{\mathcal{R}}_{N,w}(h) \leq \left(1 - \frac{\Lambda}{\Lambda_\star}\right)^2 \frac{\|\mathbf{y}\|_2^2}{N}. \quad (22)$$

If $\mathbf{y} = 0$, the zero function interpolates with norm zero.

What this establishes. Once the feature vectors are distinct, a finite readout norm suffices to interpolate their labels, and scaling the interpolant gives a constructive training-risk upper bound below that norm.

What this does not establish. Distinctness is an assumption, not a consequence of Proposition 2 and Lemma 3. The theorem does not establish feature-map injectivity, favorable conditioning of K , good test risk, or a property unavailable to a comparable classical feature map.

5.6 Beta mixing of strided windows

Lemma 8 (Mixing coefficient of the window process). *Let W be the process in Equation (1) with stride $s = w + g$. For every integer $a \geq 1$,*

$$\beta_W(a) \leq \beta_Z(as - w). \quad (23)$$

In particular, $\beta_W(1) \leq \beta_Z(g)$.

The bound is conservative by one time index under Equation (8): the first raw input in a future window at window lag a occurs at $as - w + 1$, and monotonicity of beta coefficients yields Equation (23).

What this establishes. Dependence between window observations can be controlled by the mixing of the underlying input-output process at the actual raw-time separation.

What this does not establish. Nonoverlapping or gapped windows are not independent unless the corresponding mixing coefficient is zero.

5.7 Generalization with dependent windows

For a sample $\mathcal{S}_\mu = (\widetilde{W}_1, \dots, \widetilde{W}_\mu)$, define the empirical Rademacher complexity using the convention

$$\widehat{\mathfrak{R}}_{\mathcal{S}_\mu}(\mathcal{F}) := \mathbb{E}_\sigma \left[\sup_{f \in \mathcal{F}} \frac{2}{\mu} \sum_{j=1}^{\mu} \sigma_j f(\widetilde{W}_j) \right], \quad \sigma_j \stackrel{\text{i.i.d.}}{\sim} \text{Unif}\{-1, +1\}. \quad (24)$$

For a length- $N = 2\mu a$ sequence of windows, $\mathcal{S}_\mu$ denotes one fixed position from each of the μ odd blocks of length a , as in the independent-block construction of [Mohri & Rostamizadeh \(2008\)](#). Let

$$\mathcal{F}_\Lambda := \{(x_{-w+1:0}, y) \mapsto (h(\Phi_w(x_{-w+1:0})) - y)^2 : h \in \mathcal{H}_{\kappa, \Lambda}\}.$$

Theorem 9 (One-sided risk bound for beta-mixing windows). *Suppose assumption 1 holds and Z is stationary and beta mixing. Let $N = 2\mu a$, with $\mu, a \in \mathbb{N}$, and set*

$$\delta' := \delta - 4(\mu - 1)\beta_Z(as - w). \quad (25)$$

For every $\delta \in (0, 1)$ satisfying $\delta' > 0$, with probability at least $1 - \delta$, simultaneously for all $h \in \mathcal{H}_{\kappa, \Lambda}$,

$$\mathcal{R}_\infty(h) \leq \widehat{\mathcal{R}}_{N,w}(h) + \widehat{\mathfrak{R}}_{\mathcal{S}_\mu}(\mathcal{F}_\Lambda) + 3(\Lambda + Y)^2 \sqrt{\frac{\log(4/\delta')}{2\mu}} + \tau_w. \quad (26)$$

Corollary 10 (Closed-form RKHS bound). *Under the conditions of Theorem 9, with the same probability, simultaneously for all $h \in \mathcal{H}_{\kappa, \Lambda}$,*

$$\begin{aligned} \mathcal{R}_\infty(h) \leq & \widehat{\mathcal{R}}_{N,w}(h) + \frac{4\Lambda(\Lambda + Y)}{\sqrt{\mu}} + 3(\Lambda + Y)^2 \sqrt{\frac{\log(4/\delta')}{2\mu}} \\ & + \frac{4\Lambda(\Lambda + Y)}{\xi} \sqrt{\frac{\nu|\mathcal{O}|}{\nu - 1} \sum_{r=1}^R \lambda_r^{2w}}. \end{aligned} \quad (27)$$

What this establishes. The theorem upper-bounds infinite-history true risk by finite-window training risk, a complexity term based on $\mu = N/(2a)$ approximately independent blocks, a confidence term, and an explicit window approximation term. It is uniform over the fixed-design RKHS ball and therefore applies to the empirical-risk minimizer within that ball.

What this does not establish. This is not a two-sided absolute generalization-gap bound. Increasing N with $a = 1$ and fixed g can violate $\delta' > 0$, so the theorem does not justify a fixed-design $1/\sqrt{N}$ claim. It also does not cover data-dependent model selection, finite-shot training features, nonstationary data, unbounded labels, excess risk relative to a Bayes predictor, or quantum advantage.

Corollary 11 (Geometrically mixing case). *Suppose, in addition, that $\beta_Z(q) \leq \beta_0 e^{-cq}$ for constants $\beta_0, c > 0$, and keep w and s fixed. For a compatible sample size $N = 2\mu a$, choose*

$$a \geq \left\lceil \max \left\{ 1, \frac{1}{cs} \log \left(\frac{4\beta_0 e^{cw} N}{\delta} \right) \right\} \right\rceil. \quad (28)$$

Then $\delta' \geq \delta/2$. For fixed $\Lambda, Y, \delta, \beta_0, c, w, s$, the two stochastic terms in Equation (27) are

$$O\left(\sqrt{\frac{\log(N/\delta)}{N}}\right).$$

If N is not compatible with the chosen a , the result applies after discarding at most $2a - 1$ terminal observations.

The truncation term τ_w remains in this corollary. A consistency claim would require a joint schedule for w , s , a , and the readout class; such a result is not asserted here.

5.8 Finite-shot local-observable estimation

Proposition 12 (Local-Pauli classical-shadow requirement). *Suppose $\mathcal{O}$ contains $M = |\mathcal{O}|$ Pauli strings of weight at most k . Under independent local random Pauli measurements, there is a universal constant $C_{\text{sh}} > 0$ such that*

$$S \geq C_{\text{sh}} 3^k \varepsilon^{-2} \log \left(\frac{2M}{\delta_{\text{sh}}} \right) \quad (29)$$

snapshots of one reservoir state suffice for simultaneous additive error at most ε for all observables, with probability at least $1 - \delta_{\text{sh}}$.

For N windows and R separately prepared sub-reservoir states, replacing δ_{sh} by $\delta_{\text{tot}}/(NR)$ gives a simultaneous guarantee over all exact feature coordinates. This requires S snapshots per state and NRS state preparations in total.

Corollary 13 (Stability of a fixed readout to measurement error). *Suppose every coordinate of $\hat{\Phi}_w - \Phi_w$ has magnitude at most ε . For every fixed $h \in \mathcal{H}_{\kappa, \Lambda}$,*

$$\left| h(\hat{\Phi}_w) - h(\Phi_w) \right| \leq \frac{\Lambda}{\xi} \sqrt{\frac{\nu R |\mathcal{O}|}{\nu - 1}} \varepsilon, \quad (30)$$

$$\left| \ell(h(\hat{\Phi}_w), y) - \ell(h(\Phi_w), y) \right| \leq \frac{2\Lambda(\Lambda + Y)}{\xi} \sqrt{\frac{\nu R |\mathcal{O}|}{\nu - 1}} \varepsilon \quad (31)$$

whenever $|y| \leq Y$.

What this establishes. Locality controls the shot exponent for the stated measurement ensemble, and uniform coordinate error has an explicit effect on evaluating a fixed readout.

What this does not establish. The result does not analyze a readout retrained on noisy features, coherent gate error, reset error, decoherence, correlated shots, or device execution. Finite-shot simulation is therefore not real-hardware validation.

6 Empirical Evaluation

Draft placeholder. The empirical section is intentionally deferred in this theory-rewrite version. Before submission, it will be populated with the audited experiment package: controlled mechanism validation, matched classical and component controls, real-world external validation, resource ablations, finite-shot simulation, uncertainty across seeds, and complete result provenance. No empirical conclusion is drawn in the present draft.

7 Discussion and Limitations

What is specific to the quantum reservoir. The state update, its reset-channel implementation, and the observable measurement process are quantum. The statistical readout and the JL preprocessing are classical. The theory proves useful structural properties of this complete construction, but it does not isolate a predictive benefit that cannot be reproduced by a classical nonlinear reservoir or random feature map. The appropriate comparative question is therefore empirical and must use matched feature, readout, split, and model-selection protocols.

Fixed-design learning theory. Theorem 9 is uniform over an RKHS norm ball after the projection, reservoirs, observable family, window scheme, and kernel hyperparameters have been fixed independently of the analyzed sample. If these choices are selected on training or validation data, the theorem must be combined with a separate model-selection argument. Bounded labels and stationary beta mixing are substantive assumptions; nonstationarity, heavy-tailed outcomes, and online adaptation are outside the current result.

Effective sample size. The independent-block parameter a is not a cosmetic proof variable. Increasing a weakens the coupling penalty through $\beta_Z(as - w)$ but decreases the effective block count $\mu = N/(2a)$. Consequently, the theory does not predict a universal $1/\sqrt{N}$ curve for a fixed gap. Under geometric mixing, the explicit block schedule in Corollary 11 yields a logarithmic degradation from the independent rate.

Window approximation. Contraction produces a transparent finite-window error, but it also imposes a memory trade-off. Stronger contraction decreases the worst-case effect of the remote past; it may simultaneously suppress task-relevant history. The bound does not select w or λ_r , nor does it imply monotonic predictive improvement as either parameter changes.

Measurement and hardware scope. The generalization theorem uses exact expectations. The shadow result covers finite sampling under a particular randomized local-Pauli measurement model, and Corollary 13 concerns evaluation of a fixed readout. Neither result covers training instability from noisy Gram matrices, gate noise, reset error, decoherence, calibration drift, or correlated device measurements. The reset circuit in Figure 2 is an implementable dilation, not real-device validation.

Resource interpretation. The JL statement controls the logical encoded width for a finite point set; it does not remove the input-dimension dependence of classical projection. Sequential spatial reuse can keep peak quantum width at one reservoir plus reset ancillas, but it increases the number of executions. Local shadows reduce the dependence on the number of requested observables to a logarithm for fixed locality while retaining a 3^k dependence on locality. Finally, dense kernel ridge regression has quadratic memory and cubic training time unless an additional approximation or solver is introduced.

No advantage claim. Nothing in the interpolation or generalization result distinguishes quantum features from arbitrary fixed features satisfying the same assumptions. Accordingly, this work does not claim quantum advantage, asymptotic runtime speedup, state-of-the-art accuracy, or general superiority over classical temporal models. Its practical case must rest on transparent operating-point comparisons after the empirical section is completed.

Societal impact. The present work develops a general-purpose learning method and does not introduce a domain-specific deployment. Its principal foreseeable risks are the ordinary risks of time-series prediction—unreliable extrapolation, distribution shift, and overconfidence—together with inflated resource or advantage claims in an early-stage quantum setting. The explicit assumptions and limitations are intended to reduce those risks.

8 Conclusion

We presented QUARK, a resource-aware temporal-learning framework that combines reset-contractive quantum reservoirs, concatenated local-observable features, and a regularized Matérn kernel readout. The analysis establishes exact trace-norm contraction, geometrically decaying finite-window feature error, stability of RKHS predictions to feature perturbations, and conditional interpolation of distinct finite feature sets. For a fixed exact featurizer and stationary beta-mixing observations, the corrected independent-block analysis gives a one-sided true-risk bound whose stochastic terms depend on an effective block count and whose approximation term exposes the reservoir memory scale. A separate classical-shadow result accounts for finite-shot estimation of local Pauli features.

These statements provide a rigorous foundation for evaluating the method, not evidence of quantum advantage or classical superiority. The remaining submission work is empirical: the completed experiment artifacts must be integrated under matched protocols, with uncertainty and resource accounting, before claims about competitiveness, operating regimes, or practical plausibility can be assessed.

References

- Stephen Boyd and Leon O. Chua. Fading memory and the problem of approximating nonlinear operators with volterra series. *IEEE Transactions on Circuits and Systems*, 32(11):1150–1161, 1985. doi: 10.1109/TCS.1985.1085649.
- Jiayin Chen and Hendra I. Nurdin. Learning nonlinear input–output maps with dissipative quantum systems. *Quantum Information Processing*, 18(7), 2019. doi: 10.1007/s11128-019-2311-9.
- Jiayin Chen, Hendra I. Nurdin, and Naoki Yamamoto. Temporal information processing on noisy quantum computers. *Physical Review Applied*, 14(2):024065, 2020. doi: 10.1103/PhysRevApplied.14.024065.
- Sanjoy Dasgupta and Anupam Gupta. An elementary proof of a theorem of johnson and lindenstrauss. *Random Structures & Algorithms*, 22(1):60–65, 2003. doi: 10.1002/rsa.10073.
- Jérôme Dedecker, Paul Doukhan, Gabriel Lang, José Rafael León, Sana Louhichi, and Clémentine Prieur. *Weak Dependence: With Examples and Applications*. Lecture Notes in Statistics. Springer, 2007. doi: 10.1007/978-0-387-69952-3.
- Keisuke Fujii and Kohei Nakajima. Harnessing disordered-ensemble quantum dynamics for machine learning. *Physical Review Applied*, 8(2):024030, 2017. doi: 10.1103/PhysRevApplied.8.024030.
- Marc G. Genton. Classes of kernels for machine learning: A statistics perspective. *Journal of Machine Learning Research*, 2:299–312, 2001. URL <https://jmlr.org/papers/v2/genton01a.html>.
- Lukas Gonon, Lyudmila Grigoryeva, and Juan-Pablo Ortega. Risk bounds for reservoir computing. *Journal of Machine Learning Research*, 21(240):1–61, 2020. URL <https://jmlr.org/papers/v21/19-902.html>.
- Lyudmila Grigoryeva and Juan-Pablo Ortega. Echo state networks are universal. *Neural Networks*, 108: 495–508, 2018. doi: 10.1016/j.neunet.2018.08.025.
- Thomas Hofmann, Bernhard Schölkopf, and Alexander J. Smola. Kernel methods in machine learning. *The Annals of Statistics*, 36(3):1171–1220, 2008. doi: 10.1214/009053607000000677.
- Fangjun Hu, Saeed A. Khan, Nicholas T. Bronn, Gerasimos Angelatos, Graham E. Rowlands, Guilhem J. Ribeill, and Hakan E. Türeci. Overcoming the coherence time barrier in quantum machine learning on temporal data. *Nature Communications*, 15(1), 2024. doi: 10.1038/s41467-024-51162-7.
- Hsin-Yuan Huang, Richard Kueng, and John Preskill. Predicting many properties of a quantum system from very few measurements. *Nature Physics*, 16(10):1050–1057, 2020. doi: 10.1038/s41567-020-0932-7.
- Hsin-Yuan Huang, Michael Broughton, Masoud Mohseni, Ryan Babbush, Sergio Boixo, Hartmut Neven, and Jarrod R. McClean. Power of data in quantum machine learning. *Nature Communications*, 12(1), 2021. doi: 10.1038/s41467-021-22539-9.
- Herbert Jaeger. The “echo state” approach to analysing and training recurrent neural networks. Technical Report GMD Report 148, GMD—German National Research Center for Information Technology, 2001.
- Wolfgang Maass, Thomas Natschläger, and Henry Markram. Real-time computing without stable states: A new framework for neural computation based on perturbations. *Neural Computation*, 14(11):2531–2560, 2002. doi: 10.1162/089976602760407955.
- Rodrigo Martínez-Peña and Juan-Pablo Ortega. Quantum reservoir computing in finite dimensions. *Physical Review E*, 107(3):035306, 2023. doi: 10.1103/PhysRevE.107.035306.

- Mehryar Mohri and Afshin Rostamizadeh. Rademacher complexity bounds for non-i.i.d. processes. *Advances in Neural Information Processing Systems*, 21, 2008. URL https://proceedings.neurips.cc/paper_files/paper/2008/hash/7eacb532570ff6858afd2723755ff790-Abstract.html.
- Riccardo Molteni, Claudio Destri, and Enrico Prati. Optimization of the memory reset rate of a quantum echo-state network for time sequential tasks. *Physics Letters A*, 465:128713, 2023. doi: 10.1016/j.physleta.2023.128713.
- Pere Mujal, Rodrigo Martínez-Peña, Gian Luca Giorgi, Miguel C. Soriano, and Roberta Zambrini. Time-series quantum reservoir computing with weak and projective measurements. *npj Quantum Information*, 9(1), 2023. doi: 10.1038/s41534-023-00682-z.
- Kohei Nakajima, Keisuke Fujii, Makoto Negoro, Kosuke Mitarai, and Masahiro Kitagawa. Boosting computational power through spatial multiplexing in quantum reservoir computing. *Physical Review Applied*, 11(3):034021, 2019. doi: 10.1103/PhysRevApplied.11.034021.
- Emilio Porcu, Moreno Bevilacqua, Robert Schaback, and Chris J. Oates. The matérn model: A journey through statistics, numerical analysis and machine learning. *Statistical Science*, 39(3):469–492, 2024. doi: 10.1214/24-STS923.
- Bernhard Schölkopf and Alexander J. Smola. *Learning with Kernels: Support Vector Machines, Regularization, Optimization, and Beyond*. MIT Press, 2002. doi: 10.7551/mitpress/4175.001.0001.
- Maria Schuld and Nathan Killoran. Quantum machine learning in feature hilbert spaces. *Physical Review Letters*, 122(4):040504, 2019. doi: 10.1103/PhysRevLett.122.040504.
- David P. Woodruff. Sketching as a tool for numerical linear algebra. *Foundations and Trends in Theoretical Computer Science*, 10(1–2):1–157, 2014. doi: 10.1561/04000000060.
- Bin Yu. Rates of convergence for empirical processes of stationary mixing sequences. *The Annals of Probability*, 22(1):94–116, 1994. doi: 10.1214/aop/1176988849.

A Technical Background

A.1 Quantum states and trace norm

An n -qubit density operator is a positive semidefinite matrix of trace one on $(\mathbb{C}^2)^{\otimes n}$. The trace norm is $\|A\|_1 = \text{Tr} \sqrt{A^\dagger A}$, and the operator norm is $\|A\|_\infty$. We repeatedly use

$$|\text{Tr}[AB]| \leq \|A\|_\infty \|B\|_1,$$

unitary invariance of the trace norm, and the fact that two density operators have trace-norm distance at most two.

A.2 Absolute regularity and independent blocks

For sigma-algebras $\mathcal{A}$ and $\mathcal{B}$, their absolute-regularity coefficient can be written as

$$\beta(\mathcal{A}, \mathcal{B}) = \frac{1}{2} \sup_{\{A_i\}, \{B_j\}} \sum_{i,j} |\mathbb{P}(A_i \cap B_j) - \mathbb{P}(A_i)\mathbb{P}(B_j)|,$$

where the supremum ranges over finite measurable partitions from the two sigma-algebras. The coefficient is monotone under inclusion of either sigma-algebra.

We use the following direct specialization of the empirical Rademacher result of [Mohri & Rostamizadeh \(2008, Theorem 2\)](#). Let $(Q_j)_{j \in \mathbb{Z}}$ be stationary and beta mixing, let $\mathcal{F}$ be a fixed class of measurable functions into

$[0, B]$, and let $m = 2\mu a$. Divide $Q_1, \dots, Q_m$ into 2μ consecutive blocks of length a , and form $\mathcal{S}_\mu$ by selecting the same fixed coordinate from every odd block. If

$$\delta'_Q := \delta - 4(\mu - 1)\beta_Q(a) > 0,$$

then, with probability at least $1 - \delta$, simultaneously for every $f \in \mathcal{F}$,

$$\mathbb{E}[f(Q_0)] \leq \frac{1}{m} \sum_{j=1}^m f(Q_j) + \hat{\mathfrak{R}}_{\mathcal{S}_\mu}(\mathcal{F}) + 3B \sqrt{\frac{\log(4/\delta'_Q)}{2\mu}}. \quad (32)$$

This result is one-sided. A two-sided statement would require a separate application to the reverse deviation and a corresponding confidence allocation.

A.3 Spectral representation of the Matérn kernel

By Bochner's theorem, the normalized stationary Matérn kernel has a probability spectral measure:

$$\kappa(z, z') = \mathbb{E}_\Omega \left[e^{i\Omega^\top (z - z')} \right] = \mathbb{E}_\Omega \left[\cos(\Omega^\top (z - z')) \right]. \quad (33)$$

Under the parameterization in Equation (11), Ω follows a centered multivariate Student distribution with 2ν degrees of freedom and scale matrix $\xi^{-2}I$. For $\nu > 1$, its covariance exists and is

$$\mathbb{E}[\Omega\Omega^\top] = \frac{\nu}{(\nu - 1)\xi^2} I. \quad (34)$$

The spectral density is strictly positive on $\mathbb{R}^p$. These standard properties also give strict positive definiteness on distinct Euclidean points (Genton, 2001; Porcu et al., 2024).

A.4 Local Pauli classical shadows

For local random Pauli measurements, inversion of the one-qubit measurement channel multiplies a nonidentity Pauli factor by three. Therefore the squared shadow norm of a Pauli string with weight q is 3^q . The median-of-means guarantee of Huang et al. (2020) then yields simultaneous estimation of M weight-at-most- k Pauli expectations with $O(3^k \varepsilon^{-2} \log(M/\delta))$ independent snapshots.

B Proofs

B.1 Finite-set JL projection

Proof. For a fixed nonzero vector v , concentration of $\|\Pi v\|_2^2 / \|v\|_2^2$ around one gives

$$\mathbb{P}\left(\left|\|\Pi v\|_2^2 - \|v\|_2^2\right| > \varepsilon \|v\|_2^2\right) \leq 2e^{-c_0 \varepsilon^2 n}$$

for a universal $c_0 > 0$. Apply this inequality to the fewer than $M^2/2$ nonzero differences $u - v$ and take a union bound. Choosing $n \geq C\varepsilon^{-2} \log(M/\delta)$ for a sufficiently large universal C makes the total failure probability at most δ . This is the standard finite-set JL argument (Dasgupta & Gupta, 2003; Woodruff, 2014). $\square$

B.2 Trace-norm contraction

Proof. The reset component cancels when two outputs with the same input are subtracted:

$$T_{r,x}(\rho) - T_{r,x}(\rho') = \lambda_r V_r(x)(\rho - \rho') V_r(x)^\dagger.$$

Homogeneity and unitary invariance of the trace norm give Equation (14).

Iterating the equality for w common inputs yields

$$\left\| \rho_0^{(r)} - \rho_0'^{(r)} \right\|_1 \leq \lambda_r^w \left\| \rho_{-w}^{(r)} - \rho_{-w}'^{(r)} \right\|_1 \leq 2\lambda_r^w.$$

To construct the causal state, initialize the recursion at time $-m$ in an arbitrary density operator and call the resulting time-zero state $\rho_0^{[m]}$. If $m' > m$, regard the recursion started at $-m'$ as having some density operator at time $-m$. The last m updates are common, so

$$\left\| \rho_0^{[m']} - \rho_0^{[m]} \right\|_1 \leq 2\lambda_r^m.$$

Thus $(\rho_0^{[m]})_m$ is Cauchy in the finite-dimensional trace-norm space and converges. The bound also shows that the limit is independent of the chosen initial states. Shifting the argument in time produces a unique causal state sequence. $\square$

B.3 Finite-window feature error

Proof. For every r and $O \in \mathcal{O}$, Lemma 3 and trace duality give

$$\left| \text{Tr} \left[O(\rho_{\infty,0}^{(r)} - \rho_{w,0}^{(r)}) \right] \right| \leq \|O\|_\infty \left\| \rho_{\infty,0}^{(r)} - \rho_{w,0}^{(r)} \right\|_1 \leq 2\lambda_r^w.$$

Squaring and summing over r and O yields

$$\|\Phi_\infty - \Phi_w\|_2^2 \leq 4|\mathcal{O}| \sum_{r=1}^R \lambda_r^{2w}.$$

Taking square roots proves the first inequality. The second follows from $\lambda_r \leq \lambda_{\max}$. $\square$

B.4 Matérn RKHS section stability

Proof. The reproducing property and normalization give

$$\begin{aligned} \|\kappa(z, \cdot) - \kappa(z', \cdot)\|_{\mathcal{H}_\kappa}^2 &= \kappa(z, z) + \kappa(z', z') - 2\kappa(z, z') \\ &= 2(1 - \varphi_{\nu, \xi}(\|z - z'\|_2)). \end{aligned}$$

Let $u = z - z'$. Using the spectral representation in Equation (33), the elementary inequality $1 - \cos t \leq t^2/2$, and Equation (34),

$$\begin{aligned} 2(1 - \kappa(z, z')) &= 2\mathbb{E}[1 - \cos(\Omega^\top u)] \\ &\leq \mathbb{E}[(\Omega^\top u)^2] \\ &= u^\top \mathbb{E}[\Omega \Omega^\top] u \\ &= \frac{\nu}{(\nu - 1)\xi^2} \|u\|_2^2. \end{aligned}$$

Taking square roots proves the claim. $\square$

B.5 Prediction and loss truncation

Proof. For any $h \in \mathcal{H}_{\kappa, \Lambda}$, the reproducing property, Lemma 5, and Proposition 4 imply

$$\begin{aligned} |h(\Phi_\infty) - h(\Phi_w)| &= |\langle h, \kappa(\Phi_\infty, \cdot) - \kappa(\Phi_w, \cdot) \rangle_{\mathcal{H}_\kappa}| \\ &\leq \Lambda \|\kappa(\Phi_\infty, \cdot) - \kappa(\Phi_w, \cdot)\|_{\mathcal{H}_\kappa} \\ &\leq \frac{2\Lambda}{\xi} \sqrt{\frac{\nu|\mathcal{O}|}{\nu - 1} \sum_{r=1}^R \lambda_r^{2w}}. \end{aligned}$$

This is Equation (19).

For $u, v \in [-\Lambda, \Lambda]$ and $|y| \leq Y$,

$$|(u - y)^2 - (v - y)^2| = |u - v| |u + v - 2y| \leq 2(\Lambda + Y) |u - v|.$$

Both predictions are in $[-\Lambda, \Lambda]$ by Equation (4). Apply the last display pointwise with $u = h(\Phi_\infty)$, $v = h(\Phi_w)$, use the prediction bound above, and take expectations. This gives $|\mathcal{R}_\infty(h) - \mathcal{R}_w(h)| \leq \tau_w$. $\square$

B.6 Conditional interpolation

Proof. The Matérn spectral density is positive everywhere. For any $\mathbf{c} \in \mathbb{R}^N$, Bochner's representation gives

$$\mathbf{c}^\top K \mathbf{c} = \int_{\mathbb{R}^p} \left| \sum_{i=1}^N c_i e^{i\omega^\top z_i} \right|^2 p(\omega) d\omega.$$

When the z_i are distinct and $\mathbf{c} \neq 0$, the exponential polynomial inside the integral is not identically zero. Since $p(\omega) > 0$, the integral is strictly positive. Hence K is positive definite and invertible.

Let $\boldsymbol{\alpha} = K^{-1} \mathbf{y}$ and

$$h_\star(\cdot) = \sum_{i=1}^N \alpha_i \kappa(z_i, \cdot).$$

Then

$$(h_\star(z_1), \dots, h_\star(z_N))^\top = K \boldsymbol{\alpha} = \mathbf{y}$$

and

$$\|h_\star\|_{\mathcal{H}_\kappa}^2 = \boldsymbol{\alpha}^\top K \boldsymbol{\alpha} = \mathbf{y}^\top K^{-1} \mathbf{y} = \Lambda_\star^2.$$

Thus $h_\star$ is feasible for every $\Lambda \geq \Lambda_\star$ and has zero empirical risk.

For $0 \leq \Lambda \leq \Lambda_\star$, the function $h_\Lambda = (\Lambda/\Lambda_\star) h_\star$ is feasible and satisfies

$$h_\Lambda(z_i) - y_i = -\left(1 - \frac{\Lambda}{\Lambda_\star}\right) y_i.$$

Its empirical risk is the right-hand side of Equation (22); the minimum over the ball cannot be larger. The case $\mathbf{y} = 0$ is immediate. $\square$

B.7 Mixing of the window process

Proof. By stationarity it is enough to compare the sigma-algebra generated by $(W_j)_{j \leq 0}$ with that generated by $(W_j)_{j \geq a}$. Every raw variable occurring in a past window has time index at most zero, so

$$\sigma(W_j : j \leq 0) \subseteq \sigma(Z_t : t \leq 0).$$

The first raw input appearing in W_a has index $as - w + 1$, and all raw variables in later windows occur no earlier. Therefore

$$\sigma(W_j : j \geq a) \subseteq \sigma(Z_t : t \geq as - w + 1).$$

Monotonicity of $\beta(\cdot, \cdot)$ under sigma-algebra inclusion gives

$$\beta_W(a) \leq \beta_Z(as - w + 1) \leq \beta_Z(as - w),$$

where the last inequality uses that mixing coefficients are nonincreasing. For $a = 1$ and $s = w + g$, $as - w = g$. $\square$

B.8 One-sided dependent-data risk bound

Proof. For $h \in \mathcal{H}_{\kappa, \Lambda}$, define

$$f_h(W_j) = (h(\Phi_w(X_{js-w+1:js})) - Y_{js})^2.$$

By Equation (4) and $|Y_t| \leq Y$,

$$0 \leq f_h(W_j) \leq B, \quad B = (\Lambda + Y)^2.$$

The class $\mathcal{F}_\Lambda = \{f_h : h \in \mathcal{H}_{\kappa, \Lambda}\}$ is fixed independently of the sample by assumption 1. Apply Equation (32) to the stationary window process. By Lemma 8,

$$\beta_W(a) \leq \beta_Z(as - w),$$

so the assumed $\delta' > 0$ is sufficient for the source theorem and gives, simultaneously for all $h \in \mathcal{H}_{\kappa, \Lambda}$,

$$\mathcal{R}_w(h) \leq \widehat{\mathcal{R}}_{N,w}(h) + \widehat{\mathfrak{R}}_{\mathcal{S}_\mu}(\mathcal{F}_\Lambda) + 3(\Lambda + Y)^2 \sqrt{\frac{\log(4/\delta')}{2\mu}}.$$

Finally, Corollary 6 gives $\mathcal{R}_\infty(h) \leq \mathcal{R}_w(h) + \tau_w$. Combining the two inequalities proves Equation (26). $\square$

B.9 Closed-form RKHS complexity bound

Proof. Condition on the selected sample $\mathcal{S}_\mu = ((u_j, y_j))_{j=1}^\mu$, where $u_j = \Phi_w(x_j)$. Define

$$\psi_j(t) := (t - y_j)^2 - y_j^2.$$

On $t \in [-\Lambda, \Lambda]$, $\psi_j(0) = 0$ and ψ_j is L -Lipschitz with $L = 2(\Lambda + Y)$. The terms y_j^2 do not affect Rademacher complexity under Equation (24): they add a quantity independent of h whose expectation over σ is zero. The Rademacher contraction lemma therefore gives

$$\widehat{\mathfrak{R}}_{\mathcal{S}_\mu}(\mathcal{F}_\Lambda) \leq 2(\Lambda + Y) \widehat{\mathfrak{R}}_{\mathcal{S}_\mu}(\mathcal{G}_\Lambda). \quad (35)$$

Writing $\kappa_{u_j} = \kappa(u_j, \cdot)$, and identifying $\mathcal{H}_{\kappa, \Lambda}$ with its prediction class on the selected features, the reproducing property and Cauchy–Schwarz yield

$$\begin{aligned} \widehat{\mathfrak{R}}_{\mathcal{S}_\mu}(\mathcal{G}_\Lambda) &= \mathbb{E}_\sigma \left[\sup_{\|h\|_{\mathcal{H}_\kappa} \leq \Lambda} \frac{2}{\mu} \left\langle h, \sum_{j=1}^\mu \sigma_j \kappa_{u_j} \right\rangle_{\mathcal{H}_\kappa} \right] \\ &= \frac{2\Lambda}{\mu} \mathbb{E}_\sigma \left\| \sum_{j=1}^\mu \sigma_j \kappa_{u_j} \right\|_{\mathcal{H}_\kappa} \\ &\leq \frac{2\Lambda}{\mu} \sqrt{\mathbb{E}_\sigma \left\| \sum_{j=1}^\mu \sigma_j \kappa_{u_j} \right\|_{\mathcal{H}_\kappa}^2} \\ &= \frac{2\Lambda}{\mu} \sqrt{\sum_{j=1}^\mu \kappa(u_j, u_j)} = \frac{2\Lambda}{\sqrt{\mu}}. \end{aligned}$$

Combining this with Equation (35) gives

$$\widehat{\mathfrak{R}}_{\mathcal{S}_\mu}(\mathcal{F}_\Lambda) \leq \frac{4\Lambda(\Lambda + Y)}{\sqrt{\mu}}.$$

Substitution into Equation (26), followed by the definition of τ_w , proves Equation (27). $\square$

B.10 Geometrically mixing case

Proof. Since $\mu = N/(2a)$ and $a \geq 1$,

$$\begin{aligned} 4(\mu - 1)\beta_Z(as - w) &\leq \frac{2N}{a}\beta_0 e^{-c(as-w)} \\ &\leq 2N\beta_0 e^{cw} e^{-cas}. \end{aligned}$$

The choice in Equation (28) makes the last expression at most $\delta/2$, so $\delta' \geq \delta/2$. It also gives $a = O(\log(N/\delta))$ with the remaining quantities fixed. Since $\mu = N/(2a)$, both stochastic terms in Equation (27) are $O(\sqrt{a/N}) = O(\sqrt{\log(N/\delta)/N})$.

If $2a$ does not divide the available sample size, retain $2a\lfloor N/(2a) \rfloor$ observations. The number discarded is smaller than $2a$, and stationarity permits applying the same result to the retained consecutive segment. $\square$

B.11 Local-Pauli shadow requirement

Proof. For local random Pauli measurements, a Pauli string of weight $q \leq k$ has squared shadow norm $3^q \leq 3^k$. The median-of-means classical-shadow guarantee applied to M observables therefore gives

$$S \geq C_{\text{sh}} 3^k \varepsilon^{-2} \log(2M/\delta_{\text{sh}})$$

for a universal constant C_{sh} ; see Huang et al. (2020). For NR independently prepared reservoir states, assign failure probability $\delta_{\text{sh}} = \delta_{\text{tot}}/(NR)$ to each state and take a union bound. Each of the NR states requires S snapshots, giving NRS state preparations. $\square$

B.12 Finite-shot feature perturbations

Proof. There are $R|\mathcal{O}|$ feature coordinates, so the coordinate-wise assumption implies

$$\left\| \widehat{\Phi}_w - \Phi_w \right\|_2 \leq \sqrt{R|\mathcal{O}|} \varepsilon.$$

The reproducing property and Lemma 5 give

$$\begin{aligned} \left| h(\widehat{\Phi}_w) - h(\Phi_w) \right| &\leq \Lambda \left\| \kappa(\widehat{\Phi}_w, \cdot) - \kappa(\Phi_w, \cdot) \right\|_{\mathcal{H}_\kappa} \\ &\leq \frac{\Lambda}{\xi} \sqrt{\frac{\nu R|\mathcal{O}|}{\nu - 1}} \varepsilon. \end{aligned}$$

This proves Equation (30). Both predictions lie in $[-\Lambda, \Lambda]$. Applying the squared-loss Lipschitz inequality from the proof of Corollary 6 proves Equation (31). $\square$